8. [Maximum mark: 7]

Consider the functions f, g and h defined as follows for $t \in \mathbb{R}$.

$$f(t) = \sin(2t + 1)$$

$$g(t) = \sin(2t + 3)$$

$$h(t) = f(t) + g(t)$$

(a) Show that $h(t) = \mathrm{Im}\left(\mathrm{e}^{2ti}(\mathrm{e}^{i} + \mathrm{e}^{3i})\right)$. [2]

(b) Write $\mathrm{e}^{i} + \mathrm{e}^{3i}$ in the form $r\mathrm{e}^{i\theta}$, where $r > 0$ and $-\pi < \theta \le \pi$. [2]

(c) Hence or otherwise, write $h(t)$ in the form $p\sin(2t + q)$, where $p > 0$ and $0 < q < 2\pi$. [3]